



Predicting Stock Price Direction with LSTM: A Fusion of Financial Sentiment Analysis with Technical Indicators

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Abstract

Stock movement prediction is a crucial field of study that can help traders on the market make better trading decisions and consequently earn more profits. Deep Learning models are increasingly recognized for their ability to make predictions based on time series data. Among traders and investors, technical and sentiment analysis using indicators remains a popular approach in the stock market. In this study, we provide an approach that combines technical and sentiment analysis to generate predictor indicators, which are then used with the proposed model to predict stock movement direction. We conduct a thorough examination of historical stock data, leveraging the TA-Lib package to calculate an array of essential technical indicators. Additionally, we delve into sentiment analysis, capitalizing on sentiment indicators previously computed in a prior study. To fine-tune our models, we employ the powerful optimization tool Optuna. Our study centres on three prominent companies, AAPL, BAC, and MSFT, all of which are integral components of the NASDAQ exchange. We performed a comparative analysis between the baseline models and the proposed LSTM (Long Short-Term Memory) model, evaluating them based on accuracy and robustness measures, in order to evaluate the performance of our model. In addition, to thoroughly assess the differences in results, we devised the Wilcoxon signed rank test. Our results show that there is 17.05% average accuracy improvement by using LSTM model with both technical and sentiment indicators as compared to previous study.

Keywords Deep learning · LSTM · Stock market · Forecasting · Optuna · Adam

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1 Introduction

The stock market plays an imperative role in the economy of any state, and has a crucial position in the economic structure of the nation (Göçken et al., 2016; Cavalcante et al., 2016). The stock market is a huge platform that provides opportunities for investors, traders, and the public to earn money. However, it's important to pay attention while investing because there are always risks involved in this area. The investment values sometimes go up or down on the basis of market behaviour. Predicting the direction of future stock movement is a critical task and its crucial to guide the investors regarding investment decisions considering the markets up and down behaviour (Zhou et al., 2019). Therefore, a lot of research focused on making such type of models which can predict the future stock prices precisely. Predicting future stock price movement is acknowledged as a difficult task in the financial time series area and also in the field of machine learning (Tay and Cao, 2001). These challenges arise because of the complex behaviour of the stock market and these behaviours include non-linearity, noisy data and chaos (Zhang et al., 2017; Bezerra and Albuquerque, 2017). There are many other related elements like political instability, sudden changes in the government regulations, economics difficulties, changes occur in the stock prices and different events like conflicts or catastrophic events all these elements have an impact on the stock market behaviours (Zhong and Enke, 2017; Chen et al., 2017). To predict the stock prices accurately variety of data needed which can give a better understanding of the behaviours and trends. There are many methods some of them find links between the changes of prices like between past and future prices using history to foresee trends (Yang, 2021). Technical analysis (Shynkevich et al., 2017) is important here, extracting useful details from the original data. These details are thought to influence future stock prices based on past behaviour. Machine learning techniques are used in addition to traditional methods to understand the relationship between past data and future stock price movements.

In recent decades, the rise of large data and the instant progress of artificial intelligence (AI) have led to numerous research efforts utilizing machine learning (ML) and deep learning (DL) methods to explore the stock market. A subset of AI, machine learning, enables computers to learn from past experiences and improve their performance in specific tasks (Chollet, 2021). Previous research has shown that machine learning techniques have a significant potential for addressing the challenges of modeling and predicting financial data that is multivariate and nonlinear (Henrique et al., 2019). Machine learning approaches can be broadly divided into two categories: (1) individual base models, such as logistic regression (LR) (Ballings et al., 2015), artificial neural networks (ANN) (Kara et al., 2011), support vector machines (SVM) (Patel et al., 2015; Huang and Yen, 2019; Mallqui and Fernandes, 2019), k-nearest neighbor (KNN) (Dash and Dash, 2016), and Naive Bayes model (NB) (Dash et al., 2019), and (2) ensemble or combination of base models such as random forest (RF) (Ghosh et al., 2019; Ribeiro and Coelho, 2020), extreme gradient boosting (XGB) (Zięba et al. 2016; Carmona et al., 2019), and so on. In the context of forecasting the direction of stock price movement, various studies have applied machine learning methods like LR, ANN, SVM, and RF. These machine learning approaches, which incorporate artificial intelligence systems, learn from training data by identifying

intricate relationships between input and output variables. Subsequently, these techniques leverage the identified relationships to predict outputs for new or testing data. However, these methods can't grasp the time patterns in stock data and struggle with data that has many features. Additionally, the SVM model takes a long time to train, making it tough to handle large data sets. The RF model, which is based on decision trees, faces similar challenges when using a lot of decision trees, leading to slower processing and efficiency issues. As observed by the author, these approaches also struggle to capture the time-series nature of stock data and have limitations in dealing with data with many features (Nikou et al., 2019).

Deep learning models are able to address problems with large datasets, including noisy data, complicated patterns, and high dimensionality. These intricacies frequently pose difficulties that conventional machine learning algorithms find difficult to solve. DL is a component of artificial neural networks (ANNs), which are composed of layers of artificial neurons that are interconnected. Artificial Neural Networks are used to make sense of complex data patterns. However, the existing ANN models fall short in accuracy (Singh and Srivastava, 2017). Researchers have explored LSTM (Fischer and Krauss, 2018), RBM (Zhu et al., 2014), and ANN (Gurjar et al., 2018). Since stock prices rely on past days data, Recurrent Neural Networks are better suited for predicting them. Numerous studies have explored different neural network structures for modelling movement patterns. Some findings indicate that recurrent neural networks (RNNs) outperform other architectures (Rather et al., 2015; Tibshirani, 1996). Among various RNN variants, Long Short-Term Memory (LSTM) stands out (Graves, 2013; Hochreiter and Schmidhuber, 1997). LSTM is a powerful tool capable of learning complex patterns over extended periods, making it highly accurate in stock movement prediction. This study introduces the optimized Sequential LSTM approach for forecasting stock prices. Unlike traditional Recurrent Neural Networks (RNNs), LSTM can effectively grasp insights from past data to anticipate future trends. Regular RNNs struggle with managing long-term dependencies in data and often suffer from vanishing gradient issues. These problems are skilfully addressed by LSTM networks. They can remember relevant patterns across extended sequences, effectively capturing the data's reliance on past values for precise predictions. Notably, LSTM models grant precise control over retaining or discarding past context, thus avoiding the vanishing gradient challenge. These intrinsic benefits of LSTM networks position them as an ideal choice for enhancing the accuracy of future stock price predictions.

Various indicators have been employed to forecast stock movements, with the prevalent ones encompassing technical indicators, financial text data, and time series stock prices. Technical analysis focuses on picking technical indicators with the best predictive ability through feature engineering. Many researchers' strategies centre around stock prices and mathematical indicators such as RSI, MACD, and Bollinger Bands. The author use linear techniques like the Box-Jenkins Auto-Regressive Integrated Moving Average (ARIMA) and extract technical patterns from Candlestick charts in order to perform time series analysis (Box, 2013), a prevalent model for time series forecasting. In later stages, it was discovered that understanding market behaviour might be aided by the Hurst exponent, a statistical tool for categorizing time series. Notably, Qian and Rasheed (Qian and Rasheed, 2004) discovered that a

measure of trend prediction may be obtained by computing the Hurst exponent on price values. As different machine learning methods advance, more researchers are turning their attention to fundamental analysis. This method deals with unorganized data, often from sources like financial reports, official documents, and even online news and discussions (Nassirtoussi et al., 2014). As natural language processing (NLP) techniques become more widespread, extensive financial text data are being examined to forecast stock movements. This includes sources like media articles (Garcia, 2013), financial news, as well as comments on social media (Renault, 2020). Analysis of sentiment typically obtains information from text and employs sentiment indicators to anticipate stock trends (Nam and Seong, 2019). In our study, we have combined technical indicators with financial sentiment indicators, leveraging both historical stock data and investor sentiments to obtain a better comprehension of the variables affecting the direction of stock movement.

Existing studies have primarily focused on enhancing prediction methods from singular perspectives, often utilizing historical data alongside either technical or sentiment indicators. In contrast, our study takes a more comprehensive approach by combining sentiment analysis with technical indicators, thereby capturing a broader range of information that reflects stock movement. Notably, only a handful of prior studies have adopted this approach. For instance, a previous study (Yang et al., 2022) did explore the combined use of sentiment and technical indicators but yielded suboptimal results, largely due to a lack of optimization. In our research, we extended the range of technical indicators employed and incorporated sentiment indicators derived from a previous study (Yang et al., 2022), which used the FinBERT model to extract textual sentiment data from large amounts of financial data. In this study, we further increased the set of technical indicators to understand more behaviour of the stock and we also optimized the model by using powerful optimizing tool which is optuna to enhance the accuracy of stock price movement.

In our study, we picked three companies for the stock movement prediction from the NASDAQ exchange. These companies are AAPL, BAC and MSFT. We predicted the next day movement of these companies and for this purpose we used the deep learning model like LSTM, and we also used some machine learning models like SVM and RF. For training these models we used data which is the combination of sentiment data and technical indicators. We used optuna to tune the hyperparameter variables. The evaluation of these models was based on metrics including Accuracy score, kappa coefficient, Specificity, and Sensitivity. Among the DL and ML models considered, LSTM stood out, showcasing superior performance across the evaluation metrics, and emerging as the optimal model in our study. Furthermore, we compared our best model with previous literature, affirming its status as the most effective choice for predicting stock market movements in the mentioned three companies. To reinforce our findings, we conducted the Wilcoxon signed-rank test, providing additional assurance that our proposed model is the optimal choice in this study.

2 Related Work

2.1 Finance Sentiment Analysis for Stock Prediction

The traditional approach to time series analysis holds the view that stock fluctuations are inherently unpredictable. However, insights from theories of behavioural finance suggest that the erratic behaviour of investors can indeed impact the fluctuations in stock prices, as observed in Ritter's work (Ritter, 2003). Particularly noteworthy is the contemporary influence of social networks, where investors frequently share their opinions and sentiments. This social interaction can impact the general attitude of other investors and consequently shape their decision-making processes. There is a relationship between stock prices and the investor sentiments, and this relationship has been indicated in previous studies. Information regarding stocks also influences the prices and some scholars have mentioned their impact in their studies like discussed in the Wang's study (Sun et al., 2017) as well as the impact of finance related news which is discussed by Vargas and their team in their work in June 26-28, 2017. These all studies highlighted the role of other factors which have an impact on stock market so considering these factors is crucial to forecast accurately. Therefore, studying and analyzing the sentiment data from different sources like news and social media is important in case of predicting stock prices. For instance, one of the studies uses technical indicators from historical data and along with they used sentiment data from Twitter their findings show that including sentiment data predicted accurately S&P 500 and DJIA indices (Porshnev et al., 2013). In another separate study, LI and their team studied to analyze the tone of news articles and they developed a new dictionary called a sentiment dictionary (Li et al., 2014). They also investigated how financial news can influence stock price returns. Furthermore, the study incorporated sentiment analysis from financial Stocktwits into various deep learning-based methods for predicting stock prices. Sentiment analysis involves uncovering the underlying feelings and emotions conveyed in language (Sohangir et al., 2018). This process of natural language processing categorizes sentences into positive, negative, or neutral sentiments. Business professionals and service-oriented companies, who depend significantly on customer feedback, frequently employ sentiment analysis. Moreover, sentiment analysis holds equal significance alongside technical analysis when it comes to assessing a company's stock price. The methods used for conducting sentiment analysis span from lexicon-based approaches to contemporary models like sequence-to-sequence and transformers. The sentiment indicators in this paper are taken from previous study and these indicators are calculated by using the pre-trained FinBERT model.

2.2 FinBERT

FinBert is an open-source NLP model specialized in financial data, outperforms other NLP methods in financial sentiment analysis. Researchers are increasingly adopting the pre-training model approach to achieve high NLP performance as computing technology advances (Peters et al., 2018). BERT, or Bidirectional Encoder Representation from Transformer, a celebrated text embedding model by Google, has

significantly impacted the NLP field, performing well in activities such as question answering, text generating, phrase classification, and others (Devlin et al., 2018).

Research shows that BERT outperforms conventional NLP models, boosting accuracy on the MultiNLI dataset by 5.6%. However, for specific NLP tasks with unique vocabularies, BERT models trained on general text may underperform. To solve this, the authors created FinBERT (BERT for Financial Text Mining), a pre-trained variation on financial data that increases its performance (Liu et al., 2021). Figure 1 shows information on FinBERT, a domain-specific BERT model for financial sentiment analysis.

BERT undergoes pre-training on a vast word dataset. This method saves time and resources for new tasks by initializing the model with pre-trained weights and fine-tuning for the specific task. BERT models are pre-trained on tasks like masked language modeling and next sentence prediction using extensive data from sources like BookCorpus and English Wikipedia, totalling over 3.5 billion words. Experimental results indicate that FinBERT surpasses BERT, achieving a 14% higher performance on the Financial Phrasebank dataset compared to the state-of-the-art method (Mohammadi and Chapon, 2020).

2.3 LSTM Neural Network

Strong temporal correlations exist in historical stock transaction data, and contextual information is often overlooked by deep learning techniques when analyzing time series data. In 1997, Hochreiter and Schmidhuber proposed the LSTM neural network (Box, 2013) as a variation of the recurrent neural network (RNN) that is capable of taking both short-term and long-term reliance into account. The LSTM model employs three gate networks which are input gate, forget gate and output gate that surpasses RNN (Sundermeyer et al., 2012). The input gate controls the inclusion of fresh information in the cell state, the forget gate decides what to discard and what to retain to address gradient vanishing, and the output gate specifies how much information to display.

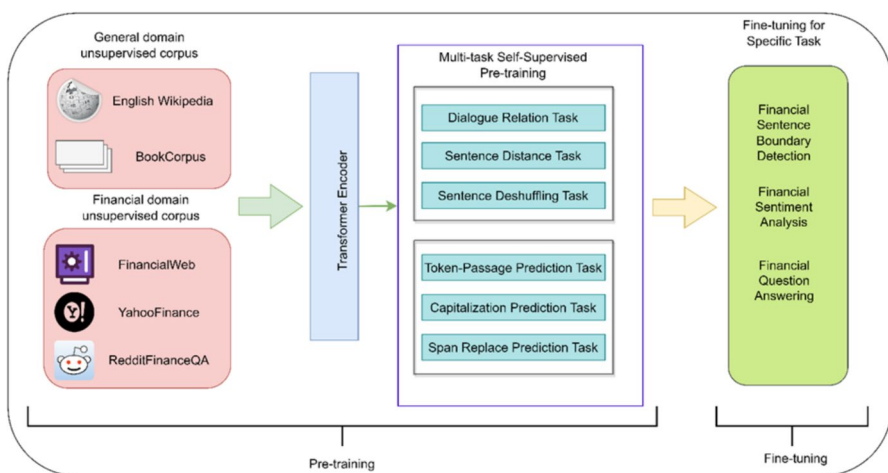


Fig. 1 The illustration of the architecture for FinBERT

Figure S1 (Supporting Information) depicts the LSTM network architecture, designed to effectively model sequence data by incorporating information from the previous state to handle the current task. Diverse applications of the LSTM network have been used for the prediction of stock market due to its efficiency in processing time series data. For example, the authors (Chen et al., 2019) employed an attention-based LSTM to forecast stock market movements and evaluated its performance in comparison to traditional LSTM. There are visual depictions of an LSTM cell, including each gate (the forget gate, input gate, and output gate), in Figs. S2, S3 and S4 (Supporting Information) along with the corresponding Eqs. S1-S5 (Supporting Information).

3 Dataset

3.1 Data Preparation

In this article, we aim to improve the accuracy of stock movement predictions by combining two critical sources of information the one source is technical indicators derived from historical transaction data and the other is sentiment indicators extracted from financial textual data. To achieve this objective, we have selected a set of companies and gathered both their transaction records and financial news data. Our analysis focuses on three representative stocks which are AAPL, MSFT, and BAC, all of which are traded on NYSE/NASDAQ.

The historical stock data for these three companies was acquired from the Yahoo Finance website, spanning the period from 2012 to 2020. This dataset contains essential attributes for each of the selected companies, such as opening and closing prices, highest and lowest prices, and trade volume.

Conversely, for the financial textual data, we have sourced sentiment indicators directly from a reference paper (Yang et al., 2022). Table 1 shows an example of textual data after cleaning, consisting of five indicators. The inclusion of the technical indicators as well as sentiment indicators allows us to provide a more comprehensive view of stock movement dynamics.

Notes

- **Newspaper_ID:** Unique identifier for each article.
- **Ticker:** Symbol representing a particular stock.
- **News Title:** Headline of the article.
- **News content:** Detailed information provided in the article.
- **Published Date:** Date when the news was published.

Table 1 The financial textual data example

Newspaper_ID	Ticker	News Title	News Content	Published Date
2060327	NIO	What's happening?	Shares of Chinese electric car maker NIO were...	2020-1-15

3.2 Technical and Sentiment Indicators

In this study, we undertook a comprehensive approach to forecast stock movements by combining both sentiment and technical indicators. For the sentiment analysis aspect, we acquired four sentiment indicators from the reference paper. These sentiment indicators were calculated using the FinBERT model, a cutting-edge financial sentiment analysis tool. These sentiment indicators provide valuable insights into the overall market sentiment and investors' perception of each company's performance. Table 2 depicts the application of processed sentiment indicators with Articles_Number denoting the number of daily article news.

In order to enrich our analysis, we integrated the sentiment indicators with technical indicators. To achieve this, we applied the TA-Lib library, which allowed us to calculate a total of 78 technical indicators for each of the three companies. The detail of these technical indicators are shown in Table S1 (Supporting information). These technical indicators are heuristic based indicators that are produced from price, volume or open interests which give the information such as price trends, momentum, and various other important factors that have an impact on the stock price movements. The data processing flow of these indicators is shown in Fig. 2. The technical indicators are calculated from the historical transaction data and for this purpose we used technical analysis library (TA-Lib), which is included in python packages and is used for research in the financial market. By using this various indicator can be calculated such as volatile indicators, momentum indicators and many others. we calculated all these

Table 2 The example of sentiment indicators (AAPL)

Date	Articles_Number	Positive	Negative	Neutral
2012-7-16	1	0.8739	0.0412	0.0847
2012-7-19	1	0.8801	0.0958	0.0240
2012-7-23	3	0.5425	0.4244	0.0330

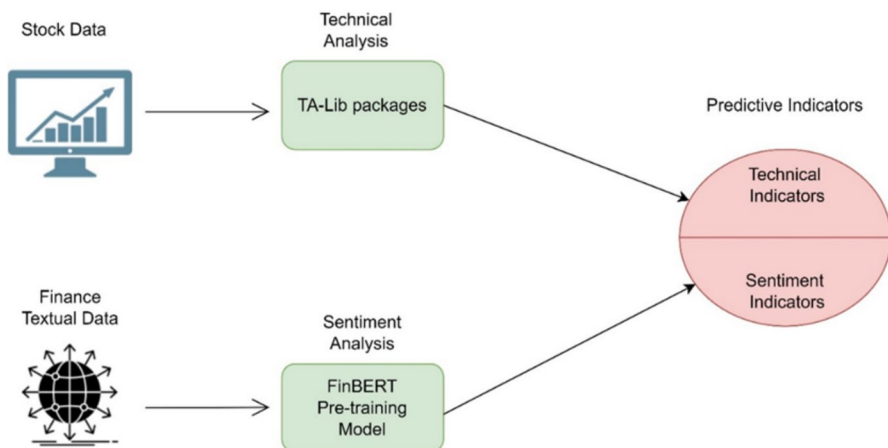


Fig. 2 The workflow for processing raw data

indicators using the historical data. The details about how to calculate these indicators can be found at this URL: <https://github.com/TA-Lib/ta-lib-python>.

3.3 Splitting

The dataset is divided into three sections which are training, validation and testing. First 70% of the data is split for training, the next 5% data is for validation and the last 25% of data is split for testing. Through this splitting we ensured that the model has enough data for training and testing.

3.4 Data Normalization

After splitting, to bring the data on a similar scale the feature scaling is applied to normalize the data. We use the StandardScaler technique to scale our data, which scales the features to have a mean of 0 and a standard deviation of 1. This feature scaling is done using the sklearn preprocessing library.

3.5 Data Reshaping

As we have timeseries data and LSTM model so it's important to arrange the data into timesteps. For this purpose, we set the value of timesteps is 30 which means that the model will use previous 30 timesteps data to predict the next data. The data used for testing, validation, and training are then modified to meet the specifications of the LSTM model. The reshaping method considers three parameters, the time step (30), the number of features (82, which are a combination of sentiment and technical indicators), and the sample size (the number of rows in the training data). This preparation ensures that the data aligns perfectly with the model, enhancing its performance and accuracy in making predictions while looking one step ahead.

4 The Proposed Model

4.1 Model Architecture

As seen in Fig. 3, our intended model is composed of two main components: data preparation and the LSTM model. Five layers make up the intended model: an LSTM layer, two dense layers, a dropout layer, and a dense output layer. By combining 78 technical indicators with 4 sentiment indicators, we are able to get the 82 predictor variables for predicting. These 82 variables were first examined for null values before being used as the LSTM's input.

The LSTM requires its data to be input in the form of time steps. We experimented with three different time step values (7, 14, and 30) while predicting one data point ahead for each company and found that a time step of 30 data points yielded the best results, as shown in Fig. 4. Therefore, we adopted a time step of 30 for our analysis. Our data is then divided into 70:5:25 ratios for training, validation, and testing. To assess the performance of the LSTM model, we employ measures such

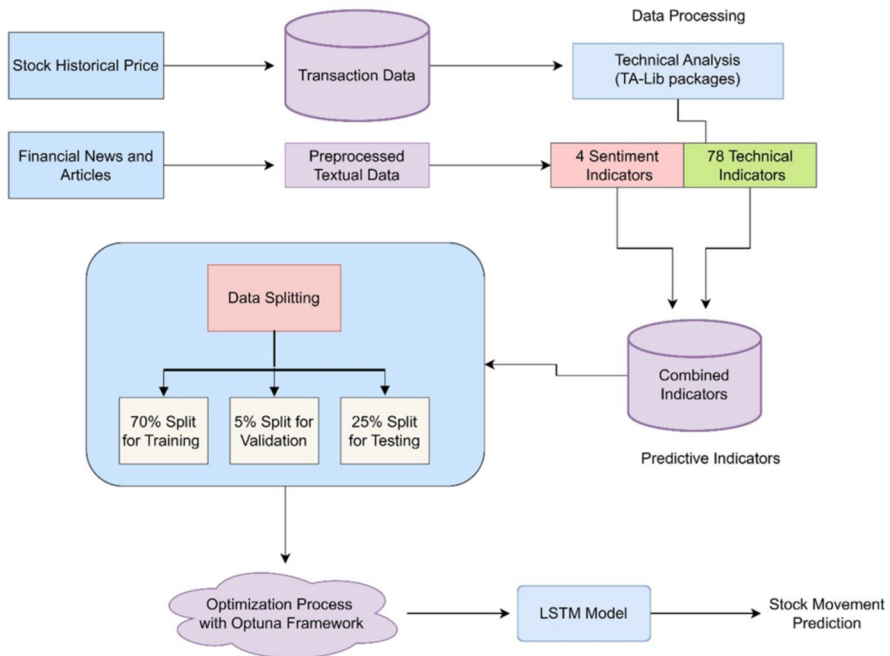


Fig. 3 The architecture of the proposed framework

as the confusion matrix, sensitivity, specificity, accuracy, ROC curve, and Kappa coefficient. The workflow of our study is depicted in Fig. 5.

4.2 Model Building

In the model building phase, we construct our LSTM model designed for stock movement prediction. This model follows a sequential architecture, allowing data to flow sequentially through its layers. The LSTM model comprises a total of 5 layers, which have been determined to yield effective predictions. The model's first layer is an LSTM taken from Kera's library. This layer's input shape is set as (30, 82) because 30 was selected as the time-step. This means that each input has 30 data points total, with 82 feature columns given as input. To keep from overfitting, a dropout rate of 0.27 is added to this layer, having 32 LSTM cells. After this, both the third and fourth layers are Dense layers with shapes 32, 47, respectively. Each layer contains an L2 regularizer with a value of 0.12 and a ReLU activation function. The fourth layer of the model is a dropout layer with a 0.35 dropout rate in order to further optimize the model's generalizability. The last one is dense layer, having a shape of (1) and uses an activation function that is sigmoid. This setup illustrates that the model's output layer is densely connected to all of the previous output layers, resulting in a single output that represents the predictions from each LSTM cell.

To minimize the loss function during the model training we used Adam optimizer with learning rate of 0.001. As we are predicting the stock movement for the next day, so our problem is a binary classification problem for this we used the binary

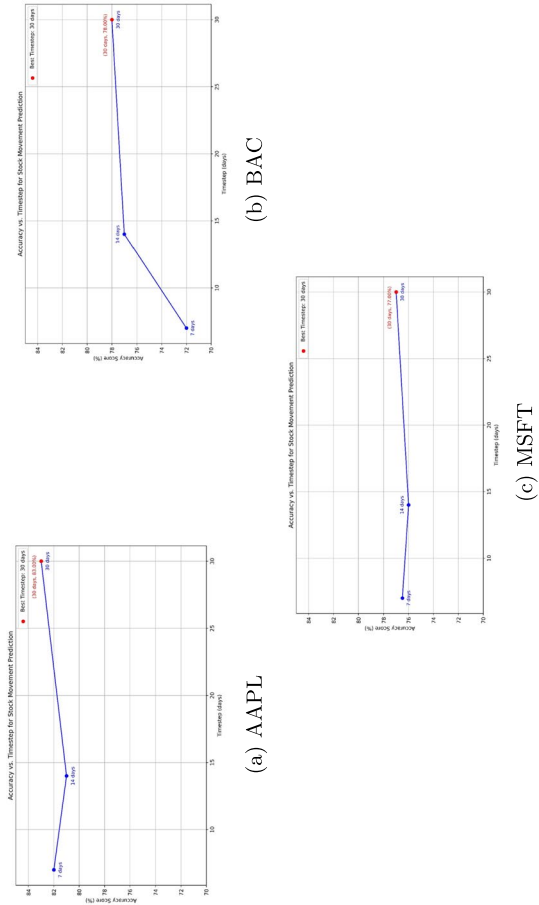


Fig. 4 Comparison of Best Timesteps for Stock Movement Prediction in Three Companies

cross-entropy loss function which basically assess how well our model is learning. We optimized the performance of our model by fine tuning the parameters and these parameters are fine-tuned by the Optuna.

4.3 Optimization

While aiming at model excellence, we used Optuna, a framework of automated hyperparameter optimization that makes the process of finding suitable configurations for machine learning models easier. Optuna helped us to pave the way through the enormous search space of hyperparameters, thus, fine-tuning our LSTM model for stock movement prediction. Optuna's method consists of formulating the objective function, which searches the hyperparameter space efficiently to maximize some evaluation metric. Optuna's method is to set an objective function which systematically traverses the hyperparameter search space to maximize a defined evaluation metric.

The hyperparameters of our optimized LSTM model include dropout (0.27, 0.35) rates, LSTM unit counts (32), L2 regularization (0.12), dense layer configurations (32,48), learning rate (0.001), and batch size (100). We set all these parameters by using optuna, we can thus easily approach the hyperparameter space and tune our model such that it will accurately predict stock price movements.

4.4 Model Compilation and Fitting

The LSTM model goes through a total of 70 epochs with a batch size of 100. To implement early stopping, a patience parameter is set to 10 which helps to lessen overfitting and improve the efficiency of the model. Besides that, 5% of the data is used for validation to check the model's ability to generalize. The model makes the predictions on the training, validation, and testing data, the accuracy measures are used to evaluate its performance.

5 Results

5.1 Evaluation Metrics

Predicting stock movement presents a binary classification problem, and an uneven sample distribution may make it difficult for accuracy metrics to assess the model's performance. To address this, we include additional evaluation metrics like the confusion matrix, sensitivity, specificity, ROC curve, and kappa coefficient offer a more thorough evaluation of the model's prediction performance. The initial metric employed in our analysis is the kappa coefficient, a statistical measure that assesses the agreement level between two annotators in a classification problem. Cohen's kappa, as it is commonly known, quantifies the extent of agreement between these two evaluators (Wan et al., 2015). Mathematically, the kappa coefficient (k) is calculated using the formula:

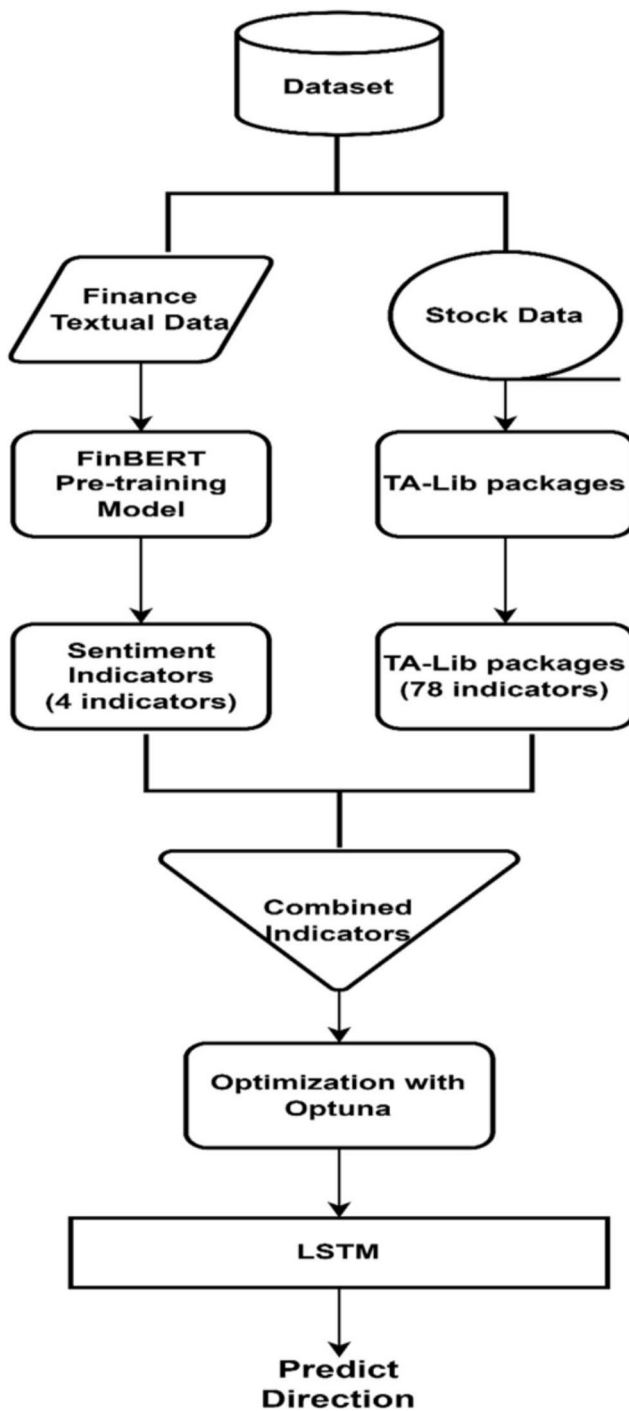


Fig. 5 Workflow of our study

$$k = \frac{(p_o - p_e)}{(1 - p_e)}$$

In this case, p_o stands for the observed agreement ratio, or empirical likelihood of agreement on the label assigned to any sample, while p_e stands for the expected agreement if both annotators assign labels at random. A pre-annotator empirical prior over the class labels is used to estimate p_e . By employing the kappa coefficient, we can gauge the level of consensus between the annotators and better understand the reliability of their classifications. This valuable metric helps us make informed decisions about the quality of the annotations and their impact on the classification outcomes.

For the purpose of evaluating the model, we can compute crucial performance metrics like accuracy, sensitivity, and specificity using the confusion matrix (Hassan et al., 2023; Gaffar et al., 2023). The Accuracy matrix calculates how accurate the model's predictions are overall. It shows the percentage of accurately predicted samples in the dataset relative to the total number of samples. A more accurate value denotes a more trustworthy model. Mathematically, the accuracy metric is calculated using the formula:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

The sensitivity (True Positive Rate) statistic shows how good the model is in detecting positive instances such as predicting the recurrence of a specific event or class accurately (Zahid et al., 2024; Akbar et al., 2025). Sensitivity metric is a mathematical approach determined using the following formula:

$$Sensitivity = \frac{TP}{TP + FN}$$

True Negative Rate (TNR) or specificity is a measure of how successful the model is in identifying negative cases, such as correctly estimating that a certain event or class is not going to happen. The specificity metric is mathematically expressed as the following formula:

$$Specificity = \frac{TN}{TN + FP}$$

Although the above metrics are helpful in knowing how the model's forecasting results are, they may not cover the whole aspect of how good the model is at predicting. In this respect, we offer ROC curves built on measurements of specificity and sensitivity. The ROC curve (sensitivity on y-axis and specificity on x-axis) shows the connection between false positive rate (FPR) and true positive rate (TPR). ROC curve is a tool for analysing the sensitivity of the model in terms of its false positive rate as well as the sensitivity of the model with respect to the true positive samples (Akbar et al., 2025). Further TPR indicating the correct identification of

positive samples is the reason why FPR value being lowered gives the impression of misclassification of negative as positive samples to a lesser extent. Besides that, the area under the ROC curve (AUC) is also used to summarize the results. The AUC value comes from finding the ROC curve at different thresholds. The AUC value is higher when the model efficiency is better. This is as a result of the model being able to differentiate between classes and make correct predictions.

5.2 Model Evaluation and Benchmark Comparison

In our study, we followed a specific data partitioning approach, utilizing 25% of the data for testing and 70% of the data for training the model. The evaluation of our proposed model was carried out on three prominent companies: AAPL, MSFT, and BAC. To establish a benchmark, we contrasted our model with five other cutting-edge models. During the training process, the model underwent 70 epochs to optimize its performance. Our findings are presented in a comprehensive Table 3, showcasing the accuracy and other metric values for each model, including our proposed one, across the three companies. Notably, our model exhibited the best performance across all metrics, indicating its potential usefulness in predicting stock movements and aiding investors in making informed decisions.

All experiments, including model training, validation, and testing, were conducted on a workstation equipped with an Intel Core i7-9700 CPU (3.00 GHz, 8 cores), 32 GB RAM, and running the Windows 10 operating system. On average, each epoch required approximately 6 seconds and the overall training process was stable with reasonable CPU and memory utilization.

Figure 6 presents the training and validation loss curves for the three datasets. The figure illustrates that both training and validation losses consistently decreased and converged smoothly over the epochs. The convergence pattern indicates no significant overfitting or divergence, suggesting the generalizability of the model in all three cases.

Table 3 Model performance in terms of predictions

Company	Model	Kappa	AUC	Sensitivity	Specificity	Accuracy
AAPL	LSTM	0.6486	0.9171	0.8302	0.8343	0.8917
	RF	0.4653	0.7184	0.9018	0.5350	0.7654
	SVM	0.3889	0.7064	0.6679	0.7452	0.6966
	LSTM_TI	0.3725	0.7356	0.5568	0.8471	0.7120
	LSTM_SI	0.0027	0.5169	0.0026	0.9836	0.6280
MSFT	LSTM	0.5963	0.9117	0.8421	0.7518	0.8743
	RF	0.4128	0.6965	0.8894	0.5036	0.7278
	SVM	0.3283	0.6503	0.9696	0.3310	0.6966
	LSTM_TI	0.2065	0.5962	0.2020	0.9123	0.6210
	LSTM_SI	0.0194	0.4955	0.9673	0.1000	0.5810
BAC	LSTM	0.5742	0.8690	0.7817	0.7952	0.8578
	RF	0.4574	0.7281	0.7658	0.6904	0.7316
	SVM	0.2632	0.6333	0.5952	0.6417	0.6298
	LSTM_TI	0.3586	0.6784	0.3115	0.9545	0.6760
	LSTM_SI	0.0036	0.4472	0.0244	1.0000	0.5455

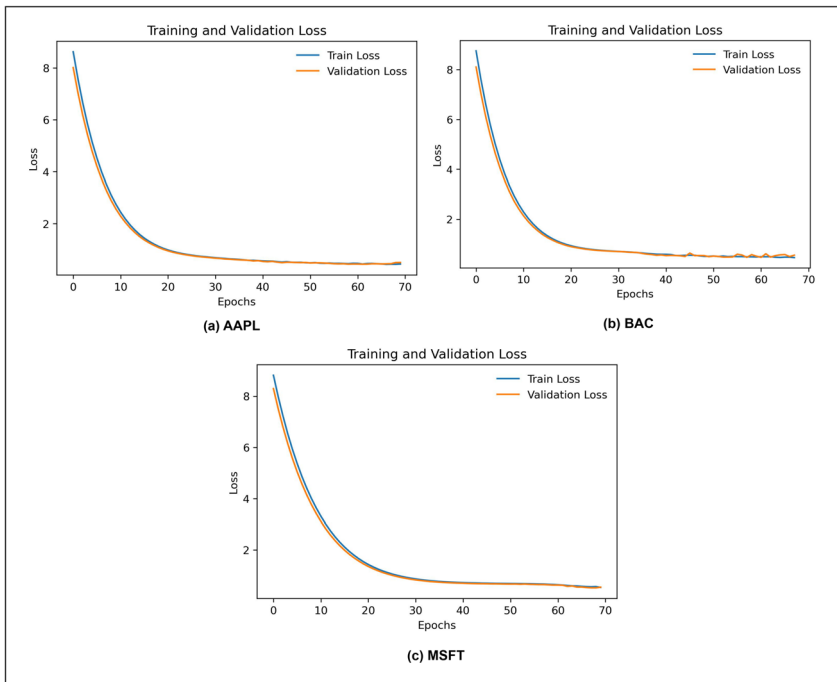


Fig. 6 Training and validation loss curves for the three stock datasets: (a) AAPL, (b) BAC, and (c) MSFT. The curves illustrate the smooth convergence behavior across epochs, demonstrating stable model learning without significant overfitting

In addition, we conducted comparisons between our proposed model and the previous study. Our proposed model consistently outperformed previous studies and classical models with regard to accuracy and AUC. The average improvement in prediction accuracy compared to a previous study was found to be almost 17%. We visualized the forecast results, as indicated in Fig. 7, based on FPR and TPR measures.

Visualizations of forecast results based on metrics for FPR and TPR demonstrated that our LSTM model boasted the largest Area Under the Curve (AUC), indicating superior performance. The ROC curves and evaluation metrics reaffirmed the enhanced capabilities of our LSTM model compared to the LASSO-LSTM, RF, and SVM models. Furthermore, compared to *LSTM_SI* (with only sentiment indicators) and *LSTM_TI* (with only technical indicators), the purposed LSTM has outperformed and demonstrated superior performance having the remarkable value of the ROC curves and performance evaluation metrics. It can be concluded that by combining technical and sentiment indicators, the suggested model is able to extract more important information.

Our proposed LSTM model, which incorporates both technical indicators and sentiment analysis, displayed superior predictive capabilities. It outperformed previous studies and classical machine learning models, highlighting the effectiveness

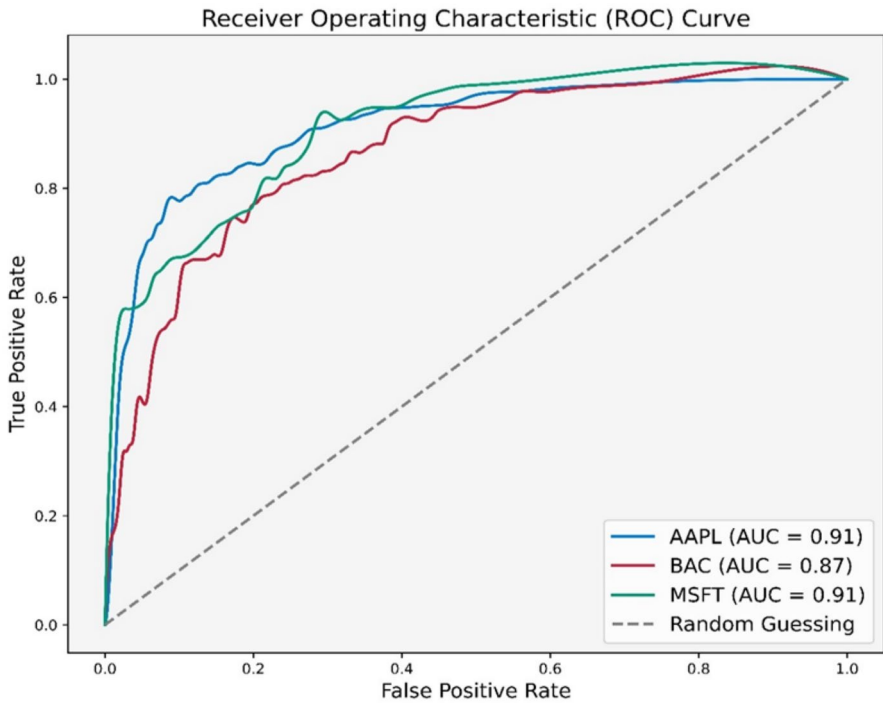


Fig. 7 ROC curves for AAPL, MSFT, and BAC on the test dataset

Table 4 Comparison with previous model

Company	Model	Kappa	AUC	Sensitivity	Specificity	Accuracy
AAPL	Our Work	0.6486	0.9171	0.8302	0.8343	0.8917
	LASSO LSTM	0.4765	0.8118	0.5674	0.8904	0.7560
MSFT	Our Work	0.5963	0.9117	0.8428	0.7518	0.8743
	LASSO LSTM	0.3988	0.7347	0.6122	0.7829	0.7160
BAC	Our Work	0.5742	0.8690	0.7817	0.7952	0.8578
	LASSO LSTM	0.5152	0.8125	0.6155	0.8836	0.7720

of combining indicators for improved prediction accuracy. The fusion of indicators in our model allowed it to capture essential information, making it a valuable tool for stock movement prediction.

5.3 Comparison with Previous Models

In the comparison with previous work, our approach using the Optuna optimization framework to fine-tune hyperparameters and train the LSTM model has yielded significant improvements in predictive performance. The resulting model showcases remarkable achievements in various evaluation metrics.

As seen in the Table 4, our model excels in predicting stock price direction, outperforming the LASSO LSTM model by a considerable margin. For the AAPL

Table 5 Wilcoxon signed rank test

Model 1	Model 2	P-value	Reject
LSTM	SVM	$7.413e^{-11}$	True
LSTM	RF	$1.513e^{-6}$	True
LSTM	LSTM_TI	$2.112e^{-6}$	True
LSTM	LSTM_SI	$1.905e^{-6}$	True
LSTM	LASSO LSTM	$2.523e^{-9}$	True

company, our model achieves a Kappa score of 0.6486, which surpasses the previous model by a significant margin, having a Kappa of 0.4765. Additionally, our model exhibits a substantially higher AUC (0.9171) compared to the LASSO LSTM model’s AUC (0.8118).

Our model further excels in terms of sensitivity, where it scores 0.8302, while the LASSO LSTM model scores 0.5674. Moreover, our model boasts a higher specificity of 0.8343, outperforming the previous model whose specificity is 0.7829. Ultimately, this translates into an accuracy of 0.8917 for our model, which is substantially better than the LASSO LSTM’s accuracy of 0.7560.

Similar superior results are observed for other companies, such as MSFT and BAC, where our approach consistently outperforms the LASSO LSTM model in all the metrics mentioned. This exceptional performance highlights the efficacy of our model in stock price direction prediction, and it underscores the significance of optimizing hyperparameters for enhanced predictive accuracy and reliability.

5.4 Model Statistics Test

The Wilcoxon signed rank test is used to see if there is a substantial difference among two models by comparing their predictive results. In this study, we use this test for the statistically reliability and to observe the difference between our LSTM model and the baseline models.

We establish two major ideas, the null hypothesis implies that there are no statistical differences between the predictions of the two models, but the alternative hypothesis implies that such differences exist.

Table 5 displays the statistical test results for our LSTM model compared to the baseline models. The two models do not differ significantly if the P-value is less than or equal to 0.05. After performing the test, we got the p-values less than 0.05, it means our model outperforms the baselines in performance. The results consistently show that our LSTM model outperforms all other models.

6 Conclusion

For anyone interested in the stock market, accurate price direction forecasting is essential knowledge. Deep learning and machine learning technologies have been extensively used in recent years to predict stock prices. Diverse research has been implemented to develop the computational models having the remarkable performance in prediction of stock trends and assist investors in generating significant returns. However, due to the different behaviour of the dataset having linear and nonlinear

relations complex nature of financial data, in such cases it is quite challenging task to build an optimal model having remarkable performance. Moreover, it is reported that many researchers typically utilized a single method to construct the computational model. This article's proposed prediction methodology makes use of technical analysis, NLP, and LSTM. To further validate our analysis, the model which only utilized technical indicators or sentiment data are less accurate in predicting the stock movement, while our proposed approach tends to utilize both technical and sentiment data for the prediction of the stock movements.

The proposed methodology in this article utilized sentiment analysis and technical indicators to construct predictor indicators, and it surpasses study that used a LASSO with LSTM model to forecast the movement of stock. One of key features of our proposed method is, its significant extension of the set of technical indicators. In contrast to the prior study, which included 37 technical indicators, our inquiry methodically covered 78 indicators. This expanded breadth not only increased the depth of our analysis but also made sure that market dynamics were examined in greater detail. Furthermore, the robustness and comparability of our technique are improved by our inclusion of the sentiment indicators used in the earlier work. The adoption of the LSTM model, as opposed to the LASSO LSTM model employed in the previous research, is noteworthy in our methodology. By using Optuna framework we optimised our model, that significantly improves its performance. Our model's forecast accuracy significantly increased after incorporating all 82 indications (78 technical and 4 sentiment indicators) and using sophisticated regularization techniques. In addition, our LSTM with a combination of indicators outperforms *LSTM_SI* and *LSTM_TI* by 22.5% and 30.99%, respectively. Overall, the sentiment analysis and technical analysis both contain valuable and useful information regarding trading signals, and that combining indicators with an optimizing model can improve prediction accuracy.

There are some shortcomings. First, we only looked into the stock on the NYSE/NASDAQ. As a result, we are unable to assess the effectiveness of the proposed approach in other countries' equity markets. In the subsequent research, we will attempt to apply the proposed model to study the stock market in other countries. Another potential direction for future research could involve integrating the LSTM with alternative neural network architectures to create hybrid predictive models. Our plan is to enhance the prediction accuracy by integrating current local and global optimizers, such as particle swarm optimization and genetic algorithms, to train the model parameters through hybrid optimization algorithms.

7 Supplementary Information

The supplementary information for this study is provided in a separate Word file. Please refer to the supplementary file for detailed information.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s10690-025-09560-4>.

Author Contributions M.F. and A.Ahmed proposed the conceptualization, methodology and contributed to original draft preparation which includes writing, figures and tables. H.T and K.T. Chong supervised and reviewed the paper along with funding acquisition.

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Data Availability The dataset used and analysed in this study are available from the corresponding author on request.

Declarations

Conflict of interest The authors state that they have no known competing financial interests or personal relationships that could have influenced the work presented in this paper.

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